KEDGE Business School EBP1 Quantitative Methods 1

Worksheet 6 — Index numbers

Exercise statements

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1 Core applications

Master the essential definitions and calculations from the course.

Core application — 20 min

1.1 Exercise 1 – Elementary indices, rebasing and properties

A service costs 80 euros in 2023, 92 euros in 2024 and 101.20 euros in 2025. Calculate the three base-100 indices, verify circularity, calculate the reverse index and verify reversibility, rebase the series to 2024=100, and interpret index 126.5 correctly.

Core application — 25 min

1.2 Extra 1 — Complete price–quantity decomposition

A purchasing department observes:

Product	p0	q0	p1	q1
A	4	100	5	90
B	10	50	11	60
C	20	20	18	30

1. Calculate the four totals Σp0q0, Σp1q0, Σp0q1 and Σp1q1.
2. Calculate the value, Laspeyres-price, Paasche-price and Fisher-price indices.
3. Interpret the gap between value and price change.

Method hint:

Complete all four cross-value columns before forming any index.

2 Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice — 35 min

2.1 Exercise 2 – Laspeyres, Paasche, Fisher and factor reversal

	p_0	q_0	p_1	q_1
Product X	4	100	5	90
Product Y	10	40	12	50

Calculate the four aggregates, Laspeyres, Paasche and Fisher price indices, the corresponding quantity indices, the value index, and verify Fisher factor reversal on a base-100 scale.

Assessment / exam practice — 45 min

2.2 Exercise 4 — Complete four-product indices — exam style

For four products, prices and quantities are observed in two periods. Base 0 = 100.

Produit / Product	p_0	q_0	p_1	q_1
A	53	20	62	25
B	41	28	52	34
C	75	35	82	38
D	25	30	38	33

1. Build the four product columns p_0q_0, p_1q_0, p_0q_1, p_1q_1 and totals.
2. Compute Laspeyres, Paasche and Fisher price and quantity indices.
3. Compute the value index and verify Fisher's factor reversal. Interpret the three indices.

3 Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice — 15 min

3.1 Exercise 3 – Chain indices and deflation

Annual link indices are 105, 108 and 97. Calculate the cumulative chain index and interpret it. Explain why annual changes must be compounded rather than added. Then deflate nominal revenue of 2.4 million euros when the price index equals 120.

Advanced practice — 25 min

3.2 Extra 2 — Rebasing, inflation and real performance

A nominal revenue series is €500k, €560k and €630k. The corresponding price index (year 1 = 100) is 100, 108 and 120.

1. Calculate real revenue in year-1 euros.
2. Calculate nominal and real growth from year 1 to year 3.
3. Rebase the price index so year 2 = 100.
4. Explain the result to a manager who claims that activity grew by 26%.

Method hint:

Deflate by dividing nominal value by index/100. Rebasing changes the reference, not the underlying inflation path.